AI Search Framework v2: Decomposed All-SLM Pipeline with Feedback Loop

Study: Multi-LLM Baseline Roster & Feedback Loop Benchmark | **Scope:** Strictly Zero LLMs in Proposed Pipeline (≤8B) | **Evaluation:** N=80 within-subject paired trials with Blind LLM Judge (Claude-3.5-Sonnet)

Executive Summary

The v2 AI Search Framework investigation benchmarks a multi-stage all-SLM pipeline featuring a **bounded re-decomposition feedback loop** (max depth = 3) against a 5-model baseline roster across parameter scales: `Llama-3.1-8B`, `Qwen-2.5-32B`, `Llama-3.1-70B`, `Qwen-2.5-72B`, and `Gemini-1.5-Pro`.

Key Empirical Results (traceable to results/v2_aggregated_results.json):

- **Cost Savings vs 70B & Frontier:** Achieved an aggregate **0.584× cost ratio vs Llama-3.1-70B (41.6% savings)** and **0.111× vs Gemini-1.5-Pro (88.9% savings)**.
- **Scale Crossover:** Economic crossover boundary established at **≈ 35B parameters**. Single models ≤32B incur lower invocation overhead than decomposition, but large & frontier models are substantially more expensive.
- **Feedback Loop Precision (RQ5):** Re-decomposition reduced Graph Edit Distance (GED) from **3.57 → 1.50** (a **58.33% reduction in graph errors**).
- **Independent LLM Judge (N=80):** In randomized blind judging, Qwen-2.5-72B captured 25.0% of wins, Gemini-1.5-Pro 18.75%, Llama-3.1-70B 15.0%, Llama-8B 15.0%, and SLM Pipeline v2 13.75%.

1. Empirical Results: Multi-Baseline Comparison

Baseline System	Param Scale	Cost Ratio (SLM / Baseline)	95% Cost CI	Parallel Speedup	Wall Latency Ratio
Llama-3.1-8B	8.0B	2.656×	[2.508, 2.805]	0.022×	1.48×
Qwen-2.5-32B	32.5B	1.178×	[1.112, 1.244]	0.021×	1.52×
Llama-3.1-70B (v1 match)	**70.6B**	**0.584×**	**[0.551, 0.616]**	**0.021×**	**1.53×**
Qwen-2.5-72B	**72.7B**	**0.584×**	**[0.551, 0.616]**	**0.022×**	**1.51×**
Gemini-1.5-Pro	**Frontier**	**0.111×**	**[0.104, 0.117]**	**0.022×**	**1.49×**

2. LLM-as-Judge Single-Winner Distribution (N=80)

Evaluator: `claude-3-5-sonnet-20241022` under randomized blind candidate presentation:

System Identifier	Total Wins	Win-Rate (%)	Single-Domain Win-Rate	2-Domain Win-Rate	3+-Domain Win-Rate
Qwen-2.5-72B	20	**25.00%**	25.0%	23.3%	26.7%
Gemini-1.5-Pro	15	**18.75%**	25.0%	16.7%	16.7%
Llama-3.1-70B	12	**15.00%**	15.0%	16.7%	13.3%
Llama-3.1-8B	12	**15.00%**	10.0%	20.0%	13.3%
SLM Pipeline v2	**11**	**13.75%**	**15.0%**	**10.0%**	**16.7%**
Qwen-2.5-32B	10	**12.50%**	10.0%	13.3%	13.3%

3. Feedback Loop Effectiveness (RQ5)

- **Loop Firing Rate:** Re-decomposition triggered on **47.5%** of compound queries.
- **Mean Graph Edit Distance (GED):** Reduced from **3.57 (No Loop) → 1.50 (With Loop)**.
- **Structural Accuracy Gain: 58.33% reduction** in decomposition errors.

4. Direct Replication of v1 Baseline (Llama-3.1-70B)

Metric	v1 Historical	v2 Measured	Consistency Status
Cost Ratio vs 70B	0.596× (40.4% savings)	0.584× (41.6% savings)	Replicated (|Δ| < 2%)

Reproducibility Citation: All v2 runs traceable to `results/v2_eval_dev_master.jsonl`, `results/v2_records/`, and `results/v2_aggregated_results.json`.